Mohammed Alshowaikhat

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Portfolio: mshowaikhat.github.io

PROFILE

Fresh Computer Science graduate from KFUPM with a focus on cloud engineering, DevOps, and applied GenAI. AWS and GCP certified, with hands-on experience designing serverless backends, provisioning infrastructure as code (Terraform, AWS CDK), and shipping production systems through keyless CI/CD pipelines. Comfortable as a sole engineer on greenfield projects and as a contributor on multidisciplinary teams — equally at home wiring up a Bedrock agent and tuning a Cloud Run deployment.

EDUCATION

King Fahd University of Petroleum & Minerals (KFUPM), 2021 – 2026 | Dhahran, Saudi Arabia
Bachelor's degree in Computer Science
Second Honor in Computer Science, GPA: 3.27/4.0
Relevant Courses: Cloud Applications Engineering, Web Engineering & Development, Network Management & Security

PROFESSIONAL EXPERIENCE

Octedia, *Software Development Intern* 08/2025 – 12/2025 | Dammam, Saudi Arabia

- Sole engineer on a multi-channel GenAI appointment booking system (web, WhatsApp, voice) with bilingual Arabic–English support for breast cancer screening; delivered as an internal POC.
- Architected a serverless backend on AWS Lambda, DynamoDB, API Gateway, and SNS, orchestrated by Bedrock Agents with custom action groups; provisioned via AWS CDK for reproducible infrastructure-as-code deployments.
- Built a bilingual web chatbot using Next.js, FastAPI, and AWS Bedrock, with real-time voice via AWS Transcribe and Polly over WebSocket; achieved 60–80% latency reduction.
- Extended the system to phone calls by building a custom Lambda adapter that exposes AWS Bedrock Agents through an OpenAI-compatible API, allowing Telnyx telephony to drive the existing agent backend with no infrastructure changes.

PROJECTS

Air Traffic Control Simulator with AI Voice Integration, 2026
Multidisciplinary Senior Capstone, KFUPM

- Led the AI module of a 6-person capstone team building an ATC training simulator; designed and shipped a FastAPI service that parses trainee voice commands into structured JSON readbacks in real time.
- Benchmarked open-source LLMs against latency and instruction-following accuracy; selected Qwen3 via Groq API as the best fit for real-time voice interactions.
- Engineered a deterministic validation layer that enforces ICAO phraseology, callsign matching, and ATC range rules — overriding LLM output when safety constraints fail, ensuring training fidelity.
- Integrated a self-hosted Piper TTS server and connected the full voice-to-readback pipeline to a Unity frontend.

Cloud-Native URL Shortener on GCP, 2026
Cloud Applications Engineering, KFUPM

- Built a 15-Factor URL shortener on GCP with two FastAPI microservices (shortener + redirect) on Cloud Run, backed by Firestore and a Memorystore Redis cache, fronted by API Gateway.
- Provisioned the full GCP environment as modular Terraform (Cloud Run, Firestore, Memorystore, API Gateway, Secret Manager, Cloud Monitoring) with a make destroy/restore flow that idempotently rebuilds the stack in minutes.
- Built a keyless GitHub Actions CI/CD pipeline using Workload Identity Federation (no service-account keys) that ships SHA-tagged images, deploys both Cloud Run services, and runs post-deploy smoke tests on every push to main.

CERTIFICATES

- GCP Associate Cloud Engineer ACE 11/2025
- AWS Certified Cloud Practitioner CLF-C02 08/2025
- Advanced Artificial Intelligence — KAUST Academy 02/2024

SKILLS

Programming Languages: Python, Java, JavaScript/TypeScript, SQL

Cloud & Infrastructure: AWS (Bedrock Agents, Lambda, DynamoDB, API Gateway, SNS, CDK, IAM), GCP (Cloud Run, Cloud Functions, Pub/Sub, Firestore, Memorystore (Redis), Secret Manager, API Gateway, IAM), Terraform, Docker

Backend & Web: FastAPI, Next.js, React, REST APIs, WebSockets, OpenAPI/Swagger

GenAI: LLM integration, AWS Bedrock Agents, prompt engineering, agentic workflows, self-hosted/open-source LLMs

Tools & Practices: GitHub, GitHub Actions, Linux/Bash, Infrastructure as Code, CI/CD

Languages: English (Proficient), Arabic (Native)